

30 km<sup>3</sup>

The most plant available water the top 60 cm of Kansas soils can hold, with the whole state at field capacity.

= 30 billion m<sup>3</sup> · = 30 trillion liters

Burj Khalifa  
828 m

Empire State  
443 m

Eiffel Tower  
330 m

Great Pyramid  
139 m



Gathered into one marble, it would stand more than 4½ times taller than the world's tallest building.

12 million

Olympic swimming pools —  
one pool = 2,500 m<sup>3</sup>

94%

of full Lake Mead, the largest man-made reservoir in the U.S.

142 mm

of a single rainfall event soaking the entire state at once

1 km · everything drawn to the same scale

## CLOSER TO HOME

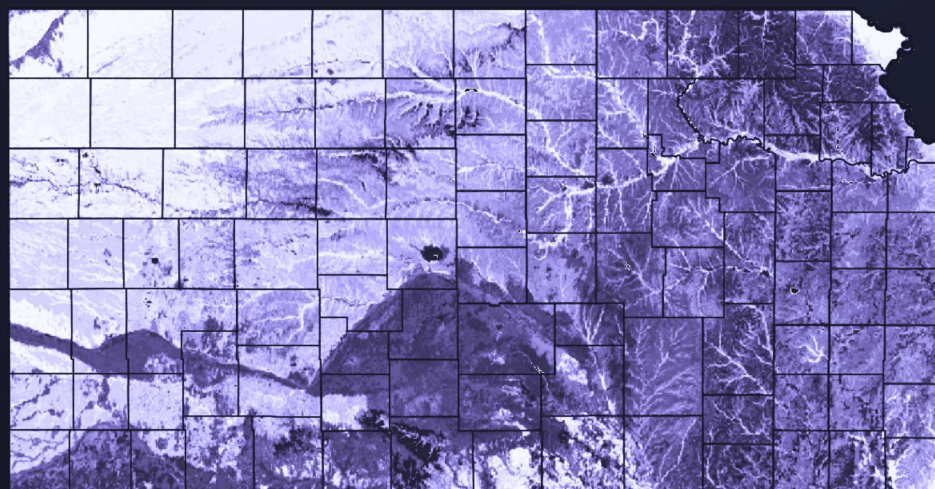
13¾ times

That's how many times this water could fill Kansas's eight largest lakes put together (≈ 2.2 km<sup>3</sup>).

Or, one lake at a time:

|              |                      |      |
|--------------|----------------------|------|
| Milford      | 0.43 km <sup>3</sup> | 70×  |
| Tuttle Creek | 0.31 km <sup>3</sup> | 96×  |
| Waconda      | 0.30 km <sup>3</sup> | 101× |
| Wilson       | 0.29 km <sup>3</sup> | 103× |
| Perry        | 0.26 km <sup>3</sup> | 116× |
| Cheney       | 0.21 km <sup>3</sup> | 145× |
| El Dorado    | 0.19 km <sup>3</sup> | 158× |
| Melvorn      | 0.19 km <sup>3</sup> | 161× |

## WHERE IT IS STORED



≤ 114 mm      ≥ 164 mm

Deep silt loams in the northwest store the most; sandy soils along the Arkansas River and in south-central Kansas store the least.

## WHY IT MATTERS

Crops, pastures, and native grasslands draw on this stored water to sustain transpiration and produce grain and forage between rains. How full this reservoir is at any moment, known as the fraction of available water, is a robust indicator of agricultural drought.

Plant available water is what a soil holds after excess water drains away (field capacity) minus what it holds too tightly for roots to extract (permanent wilting point). We report the top 60 cm, the most dynamic and densely rooted part of the soil profile, where water content increases within hours of rainfall and declines over the following days through soil evaporation and plant uptake. Both limits were estimated with custom pedotransfer functions developed by the K-State Soil Water Processes Lab, using clay and sand content maps from the USDA Natural Resources Conservation Service. Figures rounded: full Lake Mead ≈ 32 km<sup>3</sup>; Kansas land area ≈ 212,000 km<sup>2</sup>; lakes at normal pool. Created by Andres Patrignani, Soil Water Processes Lab, Agronomy, Kansas State University.